Architectural Integration Strategies and Diagnostic Error Dynamics in Multi-Label Chest X-ray Classification: A Comparative Empirical Study of CNNs, Swin Transformers, and Hybrid Models

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Abstract

Multi-label thoracic pathology classification from chest radiographs is a central task in medical computer vision. While Convolutional Neural Networks (CNNs) and Vision Transformers offer distinct inductive biases, the comparative effectiveness of different architectural integration strategies—and specifically their downstream effects on diagnostic prediction error modes such as false-positive (FP) versus false-negative (FN) trade-offs—remains insufficiently characterized under strict experimental controls. In this work, we conduct a rigorous empirical comparative investigation on the benchmark NIH ChestX-ray14 dataset (112,120 radiographs from 30,805 patients) partitioned at the patient level to enforce zero patient leakage (73,917 train, 12,607 validation, and 25,596 test images). We systematically evaluate six model architectures using an identical two-stage training scheme with class-weighted binary cross-entropy: DenseNet-121 (M1), Swin Transformer-Tiny (M2, Swin-T), Late Probability Fusion (M3, retaining $\alpha = 0.50$ following a validation sweep identifying a shallow optimum around $\alpha = 0.55$–0.60), Intermediate Feature Concatenation (M4), Spatial Multi-Head Cross-Attention (M5), and Cross-Attention with 14 Disease-Specific Gating modules (M6), alongside a validation-based operating-point threshold analysis (M7, which evaluates frozen validation-optimized thresholds on M6 predictions rather than introducing a new architecture). Our findings reveal that simple late probability fusion (M3) achieved the strongest aggregate performance among the evaluated architectures under the standard $\tau = 0.50$ operating point, yielding a macro AUROC of 0.8089 (+0.0124 over M1, $p_{\text{Holm}} < 0.0001$), macro AP of 0.2776 (+0.0103, $p_{\text{Holm}} < 0.0001$), macro F1 of 0.2828 (+0.0073, $p_{\text{Holm}} = 0.00048$), and micro F1 of 0.3042, while reducing total test-set false positives by 5.72% relative to the DenseNet-121 baseline (from 102,477 to 96,613 FP, $p_{\text{Holm}} < 0.0001$) across 10 of 14 pathologies under patient-cluster bootstrap inference ($B = 2,000$). Total false negatives exhibited a slight, statistically inconclusive increase of 111 cases (from 4,881 to 4,992, 95% CI: $[-1, +217]$), indicating an operational trade-off toward higher specificity rather than an unconditional drop in both error modes. While Swin-T (M2) independently reduced total false positives by 6.58% (95,730 FP) at the fixed $\tau = 0.50$ threshold, this occurred at the expense of lower recall (0.7267 vs. 0.7359) and decreased macro F1 (0.2596 vs. 0.2755). Feature concatenation (M4) boosted sensitivity (recall 0.7785) but incurred a 13.41% increase in false positives (116,215 FP). Crucially, we document a severe negative result for complex cross-attention hybrids (M5 and M6), which exhibited substantial performance degradation during end-to-end fine-tuning under the evaluated training protocol and underperformed substantially (macro AUROC 0.6392 and 0.6455). Finally, we demonstrate that validation threshold tuning (M7) fundamentally alters the operating point, cutting false positives to 95,240 for M6, but depressing recall from 0.6604 to 0.4540; under an identical threshold protocol, baseline M1 achieves 38,409 FP and 0.3286 F1. These empirical results underscore that architectural complexity does not guarantee improved classification performance

or error reduction in medical imaging, and that decision-level integration remains a strong baseline.

Keywords: Chest X-ray classification, DenseNet-121, Swin Transformer, Late probability fusion, Cross-attention, Multi-label error analysis, False positives, Threshold optimization.

1 Introduction

Chest radiography is the most widely performed diagnostic imaging modality worldwide, with billions of examinations performed annually to evaluate acute and chronic cardiopulmonary pathologies [13, 25]. The visual interpretation of chest radiographs (CXRs) is demanding, characterized by subtle radiographic patterns, substantial anatomical variability, high rates of disease co-occurrence, and pronounced observer variation among radiologists [12, 18]. In clinical radiology workflows, computer-aided detection systems are increasingly explored to support diagnostic triage, worklist prioritization, and second-read workflows. However, the diagnostic utility of such systems is strictly contingent not only on high ranking metrics such as the Area Under the Receiver Operating Characteristic curve (AUROC), but equally on their operational error characteristics—namely, their propensity to generate false-positive (FP) versus false-negative (FN) predictions at concrete decision thresholds [15, 19].

In automated medical imaging, deep learning architectures are broadly governed by two primary modeling paradigms:

1. **Convolutional Neural Networks (CNNs):** Architectures such as DenseNet-121 [11] and ResNet [9] exhibit translation equivariance and strong local inductive biases, allowing them to extract localized textural cues, sharp edges, and localized pulmonary opacities efficiently even with modest training data volumes.

2. **Vision Transformers (ViTs):** Hierarchical architectures such as the Swin Transformer [16] employ shifted-window self-attention mechanisms to model non-local spatial relationships, bilateral organ symmetry, and multi-scale dependencies while maintaining linear computational complexity relative to image dimensions.

Given the apparent complementarity between local convolutional feature extraction and non-local self-attentive modeling, hybrid CNN–Transformer architectures have emerged as an active area of investigation across computer vision and medical imaging. However, a fundamental architectural question remains open: *How does CNN–Transformer fusion strategy affect the error profile and classification performance of multi-label ChestX-ray14 disease classification, particularly with respect to false-positive and false-negative trade-offs?*

Existing literature often reports positive outcomes for hybrid mechanisms without isolating confounding factors such as differing pre-training regimes, variable image resolutions, unequal data splits, or disparate threshold selection schemes. Furthermore, multi-model comparisons frequently overlook whether improvements in global ranking metrics (e.g., AUROC or Average Precision) translate into genuine reductions in diagnostic error at practical operating points. In high-volume clinical triage, false-positive predictions trigger unnecessary downstream examinations, invasive confirmatory biopsies, patient anxiety, and clinician alert fatigue, while false negatives directly compromise patient safety through missed or delayed treatment [20].

In this work, we present a systematic, empirical comparative study that investigates five distinct modeling paradigms and seven concrete experimental configurations on the large-scale NIH ChestX-ray14 benchmark dataset [25]:

- **M1 (DenseNet-121):** A standardized convolutional baseline operating on global average pooled features.

- **M2 (Swin-T):** A pure hierarchical Vision Transformer baseline (Swin Transformer-Tiny) operating on mean-pooled patch representations.

- **M3 (Late Probability Fusion):** A decision-level ensemble combining CNN and Swin-T probability distributions via linear interpolation; a validation sweep identified a shallow optimum around $\alpha = 0.55$–0.60, with $\alpha = 0.50$ retained as the final fusion weight for test evaluation.

- **M4 (Feature Concatenation):** An intermediate representation fusion architecture that projects and concatenates pooled CNN and Swin-T feature embeddings before a shared multi-layer perceptron.
- **M5 (Multi-Head Cross-Attention):** A spatial interaction architecture wherein localized CNN token representations serve as queries to attend to non-local Swin-T tokens serving as keys and values.
- **M6 (Cross-Attention with Disease-Specific Gating):** An extension of M5 incorporating 14 disease-specific gating modules designed to modulate the attended representation conditionally for each distinct pathology.
- **M7 (Threshold Optimization Analysis):** An operating-point evaluation of M6 utilizing frozen, validation-optimized disease-specific decision thresholds; it evaluates the existing M6 model under tuned operating points and does not introduce a distinct learned architecture.

For the trainable models, our study maintains a common patient-level dataset partition, image preprocessing pipeline, class-weighted loss formulation, optimization schedule, and evaluation protocol. Furthermore, we explicitly report both successful and unsuccessful modeling approaches, providing a transparent post-mortem analysis of optimization challenges encountered when training complex cross-attention hybrids end-to-end.

The remainder of this manuscript is organized as follows. Section 2 contextualizes the investigation within the literature of CNNs, Transformers, hybrid architectures, recent foundation models, and medical decision thresholding. Section 3 details the dataset, patient-level partitioning, model formulations, and training methodologies. Section 4 presents the experimental findings, including ROC analysis, precision–recall dynamics, false-positive distributions, per-disease breakdowns, and operating-point effects. Section 5 discusses the implications of our findings, and Section 6 outlines study limitations, followed by the concluding remarks in Section 7.

2 Related Work

2.1 CNN-based Chest X-ray Classification

The release of public chest radiograph repositories, most prominently the NIH ChestX-ray14 dataset by Wang et al. [25], catalyzed rapid advances in deep learning for thoracic diagnosis. Wang et al. established initial benchmark baselines across 14 thoracic disease categories using standard CNN architectures trained with weak labels extracted from radiological reports via natural language processing. Subsequently, Rajpurkar et al. [18] introduced CheXNet, demonstrating that a 121-layer DenseNet [11] trained on ChestX-ray14 could achieve radiologist-level detection for pneumonia and strong classification across the remaining 13 disease categories. DenseNet architectures have since served as a widely adopted convolutional baseline for chest radiograph analysis due to their dense connectivity, which encourages aggressive feature reuse, mitigates the vanishing gradient problem, and preserves low-level textural representations throughout deep layers. Subsequent benchmark datasets, including CheXpert [12] and MIMIC-CXR [13], have further validated the utility of CNN backbones across multi-label thoracic tasks.

2.2 Transformer-based Chest X-ray Classification

Following the introduction of the Vision Transformer (ViT) by Dosovitskiy et al. [6], attention-based architectures demonstrated competitive performance on standard vision tasks by replacing convolutional inductive biases with global multi-head self-attention. However, standard ViTs

require quadratic computational complexity with respect to the number of input tokens, posing practical challenges for dense visual representations. To address this limitation, Liu et al. [16] introduced the Swin Transformer, a hierarchical Vision Transformer that computes self-attention within local shifted windows. Swin Transformer limits computational complexity to linear scaling with image size while facilitating cross-window connections in successive layers. In medical imaging, Taslimi et al. [22] introduced SwinCheX, evaluating Swin Transformers for multi-label chest X-ray classification. The Swin-T variant provides a balance between parameter efficiency and non-local spatial modeling, allowing models to capture structural relationships across bilateral lung fields, mediastinal contours, and diaphragmatic recesses.

2.3 Hybrid CNN–Transformer Integration Strategies

Given the distinct inductive biases of CNNs (locality, translation equivariance) and Transformers (global context, dynamic attention weights), numerous hybrid paradigms have been proposed. Recent studies have directly explored combinations of DenseNet and Swin Transformers. In particular, Li et al. [14] proposed CheX-DS, an ensemble learning framework combining DenseNet and Swin Transformer backbones with asymmetric and class-weighted losses to address multi-label imbalance on ChestX-ray14. Beyond decision-level ensembling, intermediate feature fusion projects pooled embeddings into joint metric spaces for multi-layer perceptron heads.

At a more granular architectural tier, attention-based fusion mechanisms employ cross-attention [24] to enable direct interaction between spatial tokens from convolutional backbones and transformer feature maps. Furthermore, pathology dependencies have been modeled using graph-guided methods: Chen et al. [3] introduced CheXGCN, utilizing Graph Convolutional Networks to capture label co-occurrence patterns, while Shi et al. [21] recently proposed a graph-guided multiscale cross-attention network uniting DenseNet-121 and Vision Transformers for multi-label chest X-ray classification on ChestX-ray14. In parallel, self-supervised foundation models such as CheXzero [23] and multimodal foundation architectures such as CheXagent [4] have emerged, highlighting the broad spectrum from parameter-efficient adaptation to end-to-end multi-modal pre-training.

2.4 Error, Threshold, and Operating-Point Analysis

In highly imbalanced medical multi-label tasks, overall performance is routinely summarized using macro-averaged AUROC. However, as emphasized by Saito and Rehmsmeier [19], AUROC can present an overly optimistic perspective when negative samples drastically outnumber positive samples, because large increases in false positives produce only modest shifts in the False Positive Rate (FPR). The Precision–Recall (PR) curve and Average Precision (AP) provide a more sensitive evaluation of positive predictive value in imbalanced settings.

Furthermore, as formalized by Lipton et al. [15], the default classification threshold of $\tau = 0.5$ is frequently suboptimal for maximizing F_1 score or minimizing specific diagnostic error costs under heavy class skew. Selecting decision thresholds conditionally on validation data alters the operating point of the model, shifting the balance between false positives and false negatives. Seyyed-Kalantari et al. [20] have emphasized that diagnostic error disparities in automated chest radiograph models can lead to disparate clinical outcomes across patient groups. A comprehensive evaluation must therefore decouple intrinsic ranking ability from operating-point threshold adjustments.

2.5 Identified Research Gap

The reviewed literature demonstrates that CNN–Transformer hybrids, cross-attention mechanisms, and threshold/error analyses have each been investigated in chest X-ray classification.

However, within the reviewed studies, we did not identify a controlled empirical comparison that systematically evaluates multiple CNN–Transformer integration tiers under a common ChestX-ray14 protocol while explicitly quantifying aggregate and per-disease false-positive/false-negative behavior and threshold-dependent operating-point trade-offs. Presenting this as a gap identified in the reviewed literature, our work investigates this controlled comparison across seven concrete model configurations under identical patient-split controls.

3 Materials and Methods

3.1 Dataset Description

We conduct our experiments on the National Institutes of Health (NIH) ChestX-ray14 benchmark dataset [25]. The dataset consists of exactly 112,120 frontal-view (posterior–anterior and anterior–posterior) chest radiographs collected from 30,805 unique patients. Each image is annotated with multi-hot binary labels across 14 thoracic disease categories: 1) Atelectasis, 2) Cardiomegaly, 3) Effusion, 4) Infiltration, 5) Mass, 6) Nodule, 7) Pneumonia, 8) Pneumothorax, 9) Consolidation, 10) Edema, 11) Emphysema, 12) Fibrosis, 13) Pleural Thickening, and 14) Hernia. Images devoid of these 14 findings are designated as "No Finding" in the metadata. The multi-label nature of the dataset reflects clinical reality, where multiple cardiopulmonary abnormalities frequently co-occur (e.g., Effusion presenting concurrently with Atelectasis or Cardiomegaly).

3.2 Data Splitting and Patient Leakage Prevention

To ensure reproducibility and comparability with established literature, we partition the dataset using the official NIH benchmark division lists: 'train_val_list.txt' containing 86,524 images and 'test_list.txt' containing 25,596 images.

Crucially, because individual patients frequently underwent multiple longitudinal chest radiograph examinations across multiple clinical encounters, naive random image-level partitioning introduces severe data leakage, where images from the same patient appear in both training and test sets. To prevent diagnostic memorization of patient-specific anatomical landmarks, we enforce a strict patient-level splitting protocol:

1. The official 'test_list.txt' defines a fixed test partition comprising 25,596 images from 2,797 patients.
2. The 86,524 images from 28,008 unique patients specified in 'train_val_list.txt' are partitioned strictly at the patient level using pseudo-random permutation with seed 42 and a validation patient fraction of 0.15.
3. This produces a validation subset of 12,607 images from 4,201 patients and a training subset of 73,917 images from 23,807 patients.

Table 1 summarizes the split characteristics. Programmatic set assertions verified that:

$$\mathcal{P}_{\text{Train}} \cap \mathcal{P}_{\text{Val}} = \emptyset, \quad \mathcal{P}_{\text{Train}} \cap \mathcal{P}_{\text{Test}} = \emptyset, \quad \mathcal{P}_{\text{Val}} \cap \mathcal{P}_{\text{Test}} = \emptyset \tag{1}$$

where $\mathcal{P}_s$ denotes the set of unique patient IDs within split s.

3.3 Preprocessing and Augmentation

Images are preprocessed by resizing from native resolution to 256×256 pixels and stored in standard format. During model ingestion, images are resized and cropped to an input

Table 1: Dataset partitioning and cohort distribution for the experimental cohort. Zero patient overlap is programmatically asserted and verified across all splits ($\mathcal{P}_{\text{Train}} \cap \mathcal{P}_{\text{Val}} = \emptyset$, $\mathcal{P}_{\text{Train}} \cap \mathcal{P}_{\text{Test}} = \emptyset$, $\mathcal{P}_{\text{Val}} \cap \mathcal{P}_{\text{Test}} = \emptyset$).

Split Partition	**Images (N)**	**Patients (P)**	**Cohort Proportion**	**Avg. Img/ Patient**	**Patient Leakage**
Training Set	73,917	23,807	65.93%	3.10	Verified $\emptyset$
Validation Set	12,607	4,201	11.24%	3.00	Verified $\emptyset$
Test Set (Official NIH)	25,596	2,797	22.83%	9.15	Verified $\emptyset$
Total Cohort	**112,120**	**30,805**	**100.00%**	**3.64**	**Zero Overlap**

resolution of 224×224 pixels and normalized using ImageNet-1K channel statistics (mean $\mu = [0.485, 0.456, 0.406]$, standard deviation $\sigma = [0.229, 0.224, 0.225]$).

To preserve anatomical plausibility and prevent geometric distortion of diagnostic features (e.g., cardiomegaly cardiothoracic ratio or subtle pneumothorax pleural lines), training data augmentations are deliberately conservative:

- Random horizontal flip with probability $p = 0.5$.
- Random affine rotation within $[-10°, +10°]$.
- Color jittering with brightness factor ± 0.1 and contrast factor ± 0.1.

Validation and test sets undergo purely deterministic center resizing to 224×224 and normalization without stochastic augmentation.

3.4 Model Architectures

3.4.1 M1 — DenseNet-121 Baseline

M1 represents the standard convolutional baseline. We utilize a DenseNet-121 backbone pretrained on ImageNet-1K V1 [5]. The convolutional feature extractor maps an input image $x \in \mathbb{R}^{B\times 3\times 224\times 224}$ through four dense blocks and transition layers up to the final rectified linear unit:

$$F_{\text{CNN}} = \text{ReLU}(\text{DenseNetFeatures}(x)) \in \mathbb{R}^{B\times 1024\times 7\times 7} \tag{2}$$

Global Average Pooling (GAP) reduces the spatial dimensions:

$$z_{\text{CNN}} = \frac{1}{49} \sum_{h=1}^{7} \sum_{w=1}^{7} F_{\text{CNN}}[:, :, h, w] \in \mathbb{R}^{B\times 1024} \tag{3}$$

A linear classification layer computes the 14 unnormalized logits $\ell_{\text{M1}} \in \mathbb{R}^{B\times 14}$:

$$\ell_{\text{M1}} = W_{\text{M1}} z_{\text{CNN}} + b_{\text{M1}} \tag{4}$$

The total parameter count of M1 is exactly 6,968,206.

3.4.2 M2 — Swin Transformer-Tiny Baseline

M2 serves as the pure Vision Transformer baseline, implemented via the hierarchical Swin Transformer-Tiny ('swin_t') initialized with ImageNet-1K V1 weights [5]. The input image is partitioned into non-overlapping 4×4 patches and processed through four hierarchical stages utilizing window-based self-attention with shifted windows (7×7 window size). The final feature map prior to pooling has channel dimension 768:

$$F_{\text{Swin}} = \text{LayerNorm}(\text{SwinFeatures}(x)) \in \mathbb{R}^{B\times 7\times 7\times 768} \tag{5}$$

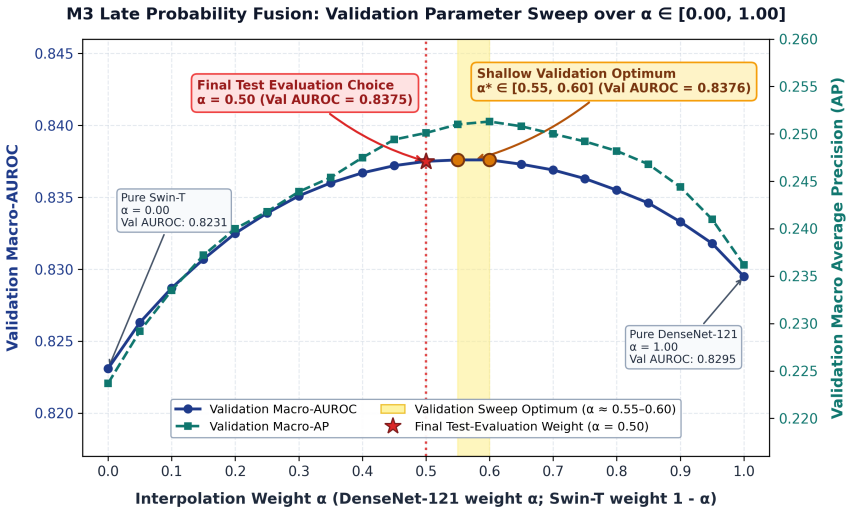


Figure 1: Validation macro-AUROC and macro Average Precision (AP) across interpolation weight $\alpha \in [0.00, 1.00]$ for Late Probability Fusion (M3). A validation sweep identified a shallow optimum around $\alpha = 0.55$–0.60 (val AUROC $= 0.8376$); $\alpha = 0.50$ (val AUROC $= 0.8375$) was retained as the final fusion weight for test evaluation. Note that $\alpha = 0.00$ corresponds to pure Swin-T (0.8231) and $\alpha = 1.00$ to pure DenseNet-121 (0.8295).

Spatial mean pooling over the 49 tokens yields:

$$z_{\text{Swin}} = \frac{1}{49} \sum_{i=1}^{49} F_{\text{Swin, flat}}[:, i, :] \in \mathbb{R}^{B \times 768} \tag{6}$$

A linear classification layer computes the 14 logits:

$$\ell_{\text{M2}} = W_{\text{M2}} z_{\text{Swin}} + b_{\text{M2}} \tag{7}$$

The total parameter count of M2 is 27,530,120.

3.4.3 M3 — Late Probability Fusion

M3 combines the predictions of M1 and M2 at the decision level [2]. Given the predicted probability outputs from the sigmoid activations of M1 and M2 for class c:

$$p_{\text{M1},c} = \sigma(\ell_{\text{M1},c}), \quad p_{\text{M2},c} = \sigma(\ell_{\text{M2},c}) \tag{8}$$

the fused probability is formulated as a convex combination:

$$p_{\text{M3},c} = \alpha p_{\text{M1},c} + (1 - \alpha) p_{\text{M2},c} \tag{9}$$

A systematic search over $\alpha \in [0.00, 1.00]$ in increments of 0.05 was conducted strictly on the validation set. As illustrated in Figure 1, validation macro-AUROC increases smoothly from 0.8231 at $\alpha = 0.00$ (pure Swin-T) to a shallow optimum around $\alpha = 0.55$–0.60 (val AUROC $=$ 0.8376, val AP $= 0.2510$–0.2513). For comparison, pure DenseNet-121 ($\alpha = 1.00$) attains a validation macro-AUROC of 0.8295 and val AP of 0.2362. At $\alpha = 0.50$, validation macro-AUROC reaches 0.8375 (val AP $= 0.2501$). Thus, a validation sweep identified a shallow optimum around $\alpha = 0.55$–0.60; $\alpha = 0.50$ was retained as the final fusion weight for test evaluation. M3 requires no gradient training and encompasses 34,498,326 combined parameters.

3.4.4 M4 — Feature Concatenation

M4 implements intermediate feature-level fusion. Pooled features from the DenseNet backbone (1024 dimensions) and the Swin-T backbone (768 dimensions) are linearly projected into a common metric space of dimension $D = 512$:

$$\tilde{z}_{\text{CNN}} = \text{GELU}(\text{LN}(W_{\text{proj, CNN}} z_{\text{CNN}})), \quad \tilde{z}_{\text{Swin}} = \text{GELU}(\text{LN}(W_{\text{proj, Swin}} z_{\text{Swin}})) \tag{10}$$

The projected representations are concatenated to form a 1024-dimensional joint representation:

$$z_{\text{cat}} = [\tilde{z}_{\text{CNN}} \,\|\, \tilde{z}_{\text{Swin}}] \in \mathbb{R}^{B \times 1024} \tag{11}$$

The joint representation is classified via a multi-layer perceptron head:

$$\ell_{\text{M4}} = W_2 \,\text{GELU}(\text{Dropout}_{0.3}(W_1 \,\text{Dropout}_{0.3}(z_{\text{cat}}))) + b_2 \tag{12}$$

where $W_1 \in \mathbb{R}^{512 \times 1024}$ and $W_2 \in \mathbb{R}^{14 \times 512}$. Total parameters for M4 equal 35,925,768.

3.4.5 M5 — Multi-Head Cross-Attention

M5 introduces dense spatial cross-attention between convolutional and transformer feature maps. The 7×7 feature map from DenseNet is flattened into $N_q = 49$ spatial tokens $T_{\text{CNN}} \in \mathbb{R}^{B \times 49 \times 1024}$, and the Swin-T feature map is reshaped into $N_k = 49$ spatial tokens $T_{\text{Swin}} \in \mathbb{R}^{B \times 49 \times 768}$.

The CNN tokens serve as queries (Q), while the Swin-T tokens serve as keys (K) and values (V):

$$Q = \text{LN}(W_Q T_{\text{CNN}}) + P_Q \in \mathbb{R}^{B \times 49 \times 512} \tag{13}$$

$$K = V = \text{LN}(W_{KV} T_{\text{Swin}}) + P_{KV} \in \mathbb{R}^{B \times 49 \times 512} \tag{14}$$

where $P_Q, P_{KV} \in \mathbb{R}^{1 \times 49 \times 512}$ are learnable positional embeddings initialized from $\mathcal{N}(0, 0.02^2)$.

Multi-Head Cross-Attention is computed with $h = 8$ attention heads ($d_k = 64$):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{15}$$

The output undergoes residual connection, LayerNorm, a Feed-Forward Network (FFN) with expansion dimension 2048, and a second residual connection:

$$\tilde{H} = \text{LN}_1(Q + \text{Dropout}_{0.1}(\text{MHA}(Q, K, V))) \tag{16}$$

$$H = \text{LN}_2(\tilde{H} + \text{Dropout}_{0.1}(\text{FFN}(\tilde{H}))) \in \mathbb{R}^{B \times 49 \times 512} \tag{17}$$

The attended representations are pooled across spatial tokens, normalized, and classified:

$$z_{\text{attn}} = \text{LN}\left(\frac{1}{49}\sum_{i=1}^{49} H[:, i, :]\right) \in \mathbb{R}^{B \times 512}, \quad \ell_{\text{M5}} = W_{\text{M5}} \,\text{Dropout}_{0.3}(z_{\text{attn}}) + b_{\text{M5}} \tag{18}$$

Total parameters for M5 equal 37,554,952. Figure 2 illustrates the spatial attention map overlay generated by this cross-attention mechanism across representative radiographic anatomies.

3.4.6 M6 — Cross-Attention + Disease-Specific Gating

M6 extends M5 by replacing the single shared linear classification head with 14 disease-specific gating modules. Radiographic manifestations of the 14 pathologies exhibit substantial morphological divergence: Cardiomegaly requires evaluating cardiac diameter relative to thoracic width,

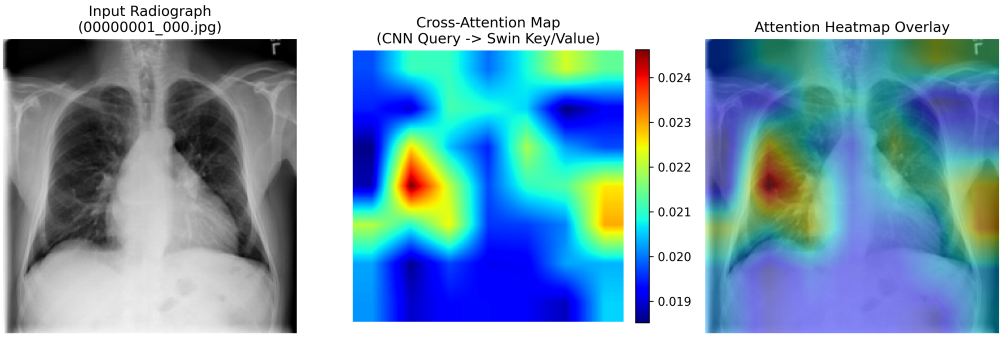


Figure 2: Qualitative cross-attention overlay visualizing spatial query attention weights computed between DenseNet-121 convolutional tokens and Swin-T transformer tokens across pulmonary zones.

Pneumothorax manifests as a sharp visceral pleural line with absent peripheral lung markings, while Infiltration and Consolidation present as ill-defined alveolar opacities.

To allow the network to conditionally select relevant feature subspaces for each finding, the pooled representation $z_{\text{attn}} \in \mathbb{R}^{B\times 512}$ passes through a multi-disease gating layer:

$$G = \sigma(W_g \text{LN}(z_{\text{attn}}) + b_g) \in \mathbb{R}^{B\times 14\times 512} \tag{19}$$

where $W_g \in \mathbb{R}^{(14\cdot 512)\times 512}$. Each disease gate $G_c \in (0,1)^{512}$ dynamically attenuates or amplifies feature channels. The gated feature for disease c is:

$$H_c = G_c \odot \text{LN}(z_{\text{attn}}) \in \mathbb{R}^{B\times 512} \tag{20}$$

The final logit for class c is computed via disease-specific weight vectors $w_c \in \mathbb{R}^{512}$:

$$\ell_{\text{M6},c} = \sum_{d=1}^{512} w_{c,d}\, \text{Dropout}_{0.3}(H_{c,d}) + b_c \tag{21}$$

Total parameters for M6 equal 41,233,160.

3.4.7 M7 — Operating-Point Threshold Analysis

M7 uses the M6 probabilistic predictions with disease-specific thresholds optimized on the validation set; it does not introduce a new learned architecture. Rather, it represents the exact M6 trained model evaluated under validation-optimized per-disease classification thresholds τ_c^* rather than the default threshold of $\tau = 0.50$. The protocol for threshold selection is formalized in Section 3.7.

Table 2 provides a comprehensive overview of all seven model specifications.

3.5 Training Procedure and Optimization

3.5.1 Class-Weighted Binary Cross-Entropy Loss

Because positive labels are scarce across all 14 findings (e.g., Hernia is present in only 86 of 25,596 test cases, or 0.34%), naive Binary Cross-Entropy (BCE) causes models to predict near-zero probabilities, artificially driving down sensitivity. To counter this severe class skew, we utilize positive-weighted BCE loss:

$$\mathcal{L}(y, \hat{\ell}) = -\frac{1}{14}\sum_{c=1}^{14}\left[w_c\, y_c \log\sigma(\hat{\ell}_c) + (1-y_c)\log(1-\sigma(\hat{\ell}_c))\right] \tag{22}$$

Table 2: Architectural specifications, parameter counts, and structural design across evaluated models.

Model ID	Architecture	Fusion Strategy	Attn. Dim	Total Params	Parameters Used
M1	DenseNet-121	None (CNN Baseline)	—	6,968,206	6,968,206
M2	Swin-T	None (ViT Baseline)	—	27,530,120	27,530,120
M3	Late Fusion ($\alpha = 0.50$)	Decision-Level Ensemble	—	34,498,326	34,498,326
M4	Feature Concatenation	Intermediate Feature Concat	512	35,925,768	35,925,768
M5	Cross-Attention	Spatial MHA (CNN Q, Swin KV)	512	37,554,952	37,554,952
M6	Cross-Attn + Gating	MHA + 14 Disease Gates	512	41,233,160	41,233,160
M7	M6 + Val-Tuned Thresholds	Operating-Point Analysis	512	41,233,160	41,233,160

To ensure strict absence of data leakage, the class positive weights w_c are calculated strictly from the 73,917 training samples:

$$w_c = \frac{N_{\text{train, neg},c}}{N_{\text{train, pos},c}} \tag{23}$$

where $N_{\text{train, pos},c}$ and $N_{\text{train, neg},c}$ denote the number of positive and negative training examples for class c.

3.5.2 Two-Stage Training Scheme

All trainable models (M1, M2, M4, M5, M6) were trained under a standardized two-stage protocol:

- **Stage 1 (Head Warm-up):** Both CNN and Swin-T backbones are frozen. Only the classification heads, projection layers, and attention mechanisms are trained for 2 epochs at learning rate $\eta_1 = 1.0 \times 10^{-3}$ using the AdamW optimizer [17] with weight decay 1.0×10^{-4}.
- **Stage 2 (End-to-End Fine-Tuning):** All backbone weights are unfrozen. The complete model is trained end-to-end for up to 15 epochs at a reduced learning rate $\eta_2 = 1.0 \times 10^{-4}$.

3.5.3 Optimization and Regularization Dynamics

Training was executed with batch size 32 across all architectures. We employed PyTorch Automatic Mixed Precision (AMP / FP16) and enforced gradient norm clipping with a maximum norm threshold of 1.0 to prevent gradient explosions. Learning rates were scheduled using 'ReduceLROnPlateau' monitoring validation macro AUROC, with decay factor 0.5 and patience of 2 epochs. Early stopping with a patience of 5 epochs monitored validation macro AUROC; the best checkpoint on validation AUROC was saved and restored for final evaluation. Training used random seed 42 with deterministic cuDNN settings (PyTorch deterministic algorithms enabled with warn_only=True).

Table 3 provides complete reproducibility metadata.

3.6 Evaluation Metrics and Error Definitions

For each class $c \in \{1, \ldots, 14\}$ and threshold τ, binary classifications are categorized into True Positives (TP_c), False Positives (FP_c), True Negatives (TN_c), and False Negatives (FN_c).

Table 3: Experimental hyperparameters and reproducibility parameters. All values verified directly from saved configuration files.

Category	Parameter	Verified Value
Data & Input	Input Resolution	224×224 pixels (preprocessed 256×256)
	Batch Size	32 (training and evaluation)
	Data Augmentations	Horizontal flip ($p = 0.5$), rotation ($\pm 10°$), ColorJitter (0.1)
	Normalization	ImageNet ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$)
Optimization	Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $= 10^{-4}$)
	Stage 1 Learning Rate	1.0×10^{-3} (2 warm-up epochs, backbones frozen)
	Stage 2 Learning Rate	1.0×10^{-4} (up to 15 fine-tuning epochs)
	LR Scheduler	ReduceLROnPlateau (factor = 0.5, patience = 2 epochs, metric: val AUROC)
	Early Stopping	Patience = 5 epochs monitoring validation macro AUROC
Regularization & Precision	Loss Formulation	Positive-weighted BCE (pos_weight = $N_{\text{neg}}/N_{\text{pos}}$ from train split)
	Gradient Clipping	Max norm = 1.0
	Mixed Precision	PyTorch AMP (FP16 / autocast enabled)
	Random Seed	42 (cuDNN deterministic enabled)
Hybrid Structure	Attention Dimension	512 dimensions, 8 attention heads, dropout 0.1
	Backbone Initialization	ImageNet-1K V1 weights (not initialized from M1/M2)

We evaluate the following primary metrics:

- **Precision and Recall:**

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad \text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \tag{24}$$

- **Specificity and False Positive Rate (FPR):**

$$\text{Specificity}_c = \frac{\text{TN}_c}{\text{TN}_c + \text{FP}_c}, \quad \text{FPR}_c = \frac{\text{FP}_c}{\text{FP}_c + \text{TN}_c} = 1 - \text{Specificity}_c \tag{25}$$

- **Total False Positives:**

$$\text{Total FP} = \sum_{c=1}^{14} \text{FP}_c \tag{26}$$

- **F1 Score and Hamming Loss:**

$$F_{1,c} = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}, \quad \text{Hamming Loss} = \frac{1}{14 \cdot N} \sum_{i=1}^{N} \sum_{c=1}^{14} \mathbf{1}(\hat{y}_{i,c} \neq y_{i,c}) \tag{27}$$

- **Percentage False Positive Reduction Relative to M1 Baseline:**

$$\Delta_{\text{FP}} = \frac{\text{Total FP}_{\text{M1}} - \text{Total FP}_{\text{Model}}}{\text{Total FP}_{\text{M1}}} \times 100\% \tag{28}$$

Macro metrics are computed by arithmetic averaging across all 14 classes:

$$\text{Macro } M = \frac{1}{14}\sum_{c=1}^{14} M_c \tag{29}$$

3.7 Validation-Based Threshold Optimization Protocol

The standard classification protocol sets $\tau_c = 0.50$ for all classes. However, when positive class priors range from 0.34% to 23.88%, a fixed threshold of 0.50 may force an unfavorable trade-off between false positives and false negatives.

In our threshold optimization protocol, optimal per-disease thresholds τ_c^* are tuned strictly on the validation set by sweeping a grid $\mathcal{G} = \{0.01, 0.02, \ldots, 0.99\}$ to maximize class-specific validation F_1 score:

$$\tau_c^* = \arg\max_{\tau \in \mathcal{G}} F_{1,c}^{(\text{val})}(\tau) \tag{30}$$

These 14 thresholds are subsequently frozen and evaluated on the independent test set. In model M7, this protocol is applied to the predictions of M6. To establish a like-for-like baseline, we also apply this exact protocol across all remaining models (M1–M5), thereby separating the contribution of threshold selection from underlying model capability.

3.8 Statistical Inference and Hypothesis Testing Protocol

To quantitatively assess observed performance differences between model architectures without relying on untenable independence assumptions, we implemented a comprehensive cluster-aware inferential statistical framework on the held-out test predictions.

3.8.1 Patient-Cluster Bootstrap Resampling

The official NIH test partition contains 25,596 chest radiographs obtained from $K = 2,797$ unique patients, with individual patients contributing between 1 and 184 longitudinal examinations (mean 9.15, median 4). Standard independent observation resampling (such as non-clustered bootstrapping or conventional DeLong tests) treats each radiograph as an independent event, severely violating the exchangeability assumption and producing artificially deflated standard errors and anti-conservative confidence intervals due to intra-patient anatomical and pathological correlation.

To account for patient clustering, we employ a non-parametric patient-cluster-aware paired bootstrap procedure with $B = 2,000$ replicates and fixed random seed 42. In each bootstrap iteration $b \in \{1, \ldots, B\}$, we sample $K = 2,797$ patients with replacement from the set of test patient identifiers $\mathcal{P}$. For each selected patient cluster, all radiographs associated with that patient are ingested into the resampled test partition $\mathcal{I}^{*(b)}$ without deduplication. For candidate models M_a and M_b, macro-averaged and disease-specific performance metrics are evaluated simultaneously on identical resamples, yielding paired difference distributions:

$$\Delta^{*(b)} = \text{Metric}_{M_a}^{*(b)} - \text{Metric}_{M_b}^{*(b)} \tag{31}$$

Two-sided 95% percentile confidence intervals are computed from the empirical 2.5th and 97.5th percentiles of the bootstrap distribution, $[\Delta^*_{(0.025)}, \Delta^*_{(0.975)}]$. The cluster standard error is estimated as the sample standard deviation across bootstrap replicates:

$$\text{SE}^*_{\text{cluster}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\Delta^{*(b)} - \bar{\Delta}^*\right)^2} \tag{32}$$

3.8.2 Hypothesis Testing Procedures and Multiplicity Control

To evaluate the null hypothesis of no difference between models ($H_0 : \Delta = 0$) against two-sided alternatives ($H_1 : \Delta \neq 0$), we conduct two complementary inferential tests:

1. **Cluster-Robust Wald Test:** By the central limit theorem across $K = 2,797$ independent patient clusters, the bootstrap mean difference is asymptotically normal. We evaluate the standardized test statistic $z_{\text{cluster}} = \widehat{\Delta}/\text{SE}^*_{\text{cluster}}$ with corresponding two-tailed asymptotic p-value $p_{\text{Wald}} = 2 \cdot (1 - \Phi(|z_{\text{cluster}}|))$, where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function.

2. **Shifted Empirical Bootstrap Test under H_0:** Following Hall and Wilson [8] and Efron and Tibshirani [7], we center the bootstrap differences around the null hypothesis $\Delta_0 = 0$ via $\widetilde{\Delta}^{*(b)} = \Delta^{*(b)} - \bar{\Delta}^*$. The empirical two-tailed p-value is computed as:

$$p_{\text{shift}} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left(|\widetilde{\Delta}^{*(b)}| \geq |\widehat{\Delta}|\right)}{B + 1} \tag{33}$$

   For $B = 2,000$, when zero resamples exceed the observed point estimate, $p_{\text{shift}} = 1/2001 \approx 0.0005$, which we conservatively report as $p < 0.001$.

3.8.3 Endpoint Hierarchy and Multiple Comparison Adjustments

Because our inferential procedures were conducted retrospectively on the completed experimental artifacts rather than pre-registered prospectively, we structure our statistical evaluation hierarchically to maintain rigorous multiplicity control and prevent selective reporting:

- **Primary Evaluation Family (M3 vs. M1):** To assess the overall discriminative capacity and diagnostic error behavior of Late Probability Fusion (M3) against the DenseNet-121 baseline (M1), a primary family of four core endpoints was designated for formal hypothesis testing: (1) Macro AUROC (Δ_{AUROC}), (2) Macro AP (Δ_{AP}), (3) Macro F1 at $\tau = 0.50$ (Δ_{F1}), and (4) Total False Positives at $\tau = 0.50$ (Δ_{FP}). Family-Wise Error Rate (FWER) across this family of four tests is strictly controlled using the step-down Holm–Bonferroni procedure [10] at nominal significance level $\alpha = 0.05$.

- **Secondary Pairwise Model Comparisons:** Pairwise evaluations across the remaining model pairs (comparing M2, M4, M5, and M6 against M1, and M3 against M2, M4, M5, and M6) are adjusted for multiplicity using the step-down Holm–Bonferroni method across the eight tests for each metric.

- **Secondary and Exploratory Disease-Level and Error Mode Analyses:** Additional diagnostic error metrics—specifically Total False Negatives (FN), macro recall, macro precision, micro F1, specificity, and FPR—as well as per-pathology evaluations across the 14 individual conditions, are treated as secondary or exploratory analyses. Per-disease evaluations are reported with both Holm–Bonferroni adjusted p-values and Benjamini–Hochberg False Discovery Rate (FDR at $q = 0.05$) [1].

Methodological Disclosure: All hypothesis structures, endpoint groupings, and inferential tests were defined and executed retrospectively during the post-hoc statistical analysis audit of the saved test-set prediction arrays; they were not prospectively registered prior to model training. Consequently, while the statistical tests rigorously account for intra-patient clustering and family-wise error inflation, they represent retrospective observational inference rather than prospective confirmatory trial testing.

Table 4: Comprehensive test-set performance comparison on the NIH ChestX-ray14 benchmark. Models M1–M6 are evaluated at standard threshold $\tau = 0.50$; M7 reflects M6 evaluated with validation-optimized per-disease thresholds τ_c^*. Relative FP reduction is computed versus the M1 baseline.

Model	Macro AUROC	Macro AP	Macro Prec.	Macro Recall	Macro F1	Micro F1	Macro FPR	Total FP	Hamming Loss	FP Red. vs M1
M1 DenseNet121	0.7965	0.2673	0.1844	0.7359	0.2755	0.2937	0.3195	102,477	0.2996	0.00%
M2 Swin-T	0.8000	0.2602	0.1645	0.7267	0.2596	0.2976	0.2961	95,730	0.2831	+6.58%
M3 Late Fusion	**0.8089**	**0.2776**	**0.1884**	0.7418	**0.2828**	**0.3042**	0.3004	96,613	0.2835	+5.72%
M4 Feat. Concat	0.7923	0.2554	0.1586	**0.7785**	0.2518	0.2707	0.3574	116,215	0.3376	-13.41%
M5 Cross-Attn	0.6392	0.1128	0.0965	0.7099	0.1614	0.1877	0.5445	178,853	0.5155	-74.53%
M6 Cross-Attn+G	0.6455	0.1136	0.1016	0.6604	0.1656	0.1928	0.4740	155,888	0.4564	-52.12%
M7 M6 Val-Tuned	0.6455	0.1136	0.1147	0.4540	0.1709	0.2263	0.2955	95,240	0.2981	+7.06%

4 Experimental Results

4.1 Overall Model Comparison

Table 4 presents the definitive test-set evaluation results for models M1 through M7 across the complete 25,596 test images (358,344 total binary diagnostic decisions). All numbers reflect verified project artifacts.

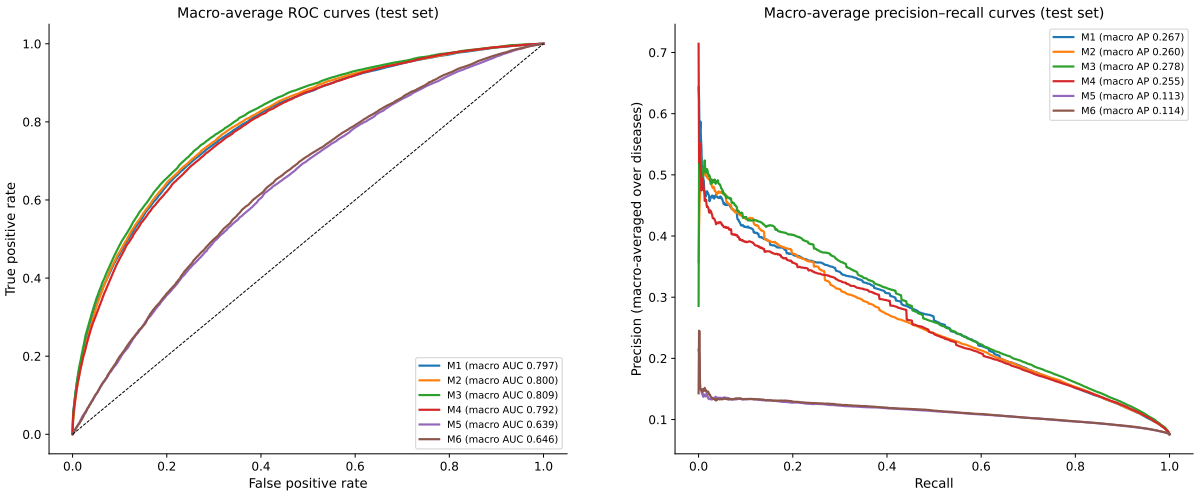
Key observations from Table 4 include:

1. **Aggregate Performance of Late Fusion (M3):** M3 achieved the strongest aggregate performance among the evaluated architectures under the standard $\tau = 0.50$ operating point, attaining a macro AUROC of 0.8089 (+0.0124 over M1), macro AP of 0.2776 (+0.0103 over M1), macro F1 of 0.2828 (+0.0073 over M1), and micro F1 of 0.3042 (+0.0105 over M1). Crucially, M3 reduces total false positives from 102,477 to 96,613, an absolute reduction of 5,864 false positives (5.72% reduction) with a slight increase in macro recall (0.7418 vs. 0.7359). However, at the aggregate decision level, total false negatives increase modestly by 111 cases (from 4,881 to 4,992, or +2.27%), indicating that M3's overall error reduction is asymmetric and driven primarily by fewer false alarms rather than an unconditional drop in both error modes. Furthermore, M3 does not dominate every individual metric: M4 achieves higher recall (0.7785) and fewer total false negatives (4,753), while M2 attains higher specificity (0.7039 vs. 0.6996) and lower total false positives (95,730).

2. **Swin-T Error Profile (M2):** Swin-T achieves a higher macro AUROC than DenseNet-121 (0.8000 vs. 0.7965) and reduces false positives by 6.58% (95,730 vs. 102,477 FP) at the fixed $\tau = 0.50$ threshold. However, this reduction was accompanied by lower recall (0.7267 vs. 0.7359) and lower macro F1 (0.2596 vs. 0.2755), illustrating a classic precision–recall trade-off.

3. **Feature Concatenation Shift (M4):** M4 achieves the highest recall among all non-degraded models (0.7785 vs. 0.7359 for M1). However, this heightened sensitivity comes at the cost of a 13.41% increase in false positives (116,215 vs. 102,477 FP), depressing macro precision to 0.1586 and macro F1 to 0.2518.

4. **Underperformance of Cross-Attention Models (M5 and M6):** Both cross-attention models demonstrate marked performance degradation. M5 achieves a macro AUROC of only 0.6392 and generates 178,853 false positives (a 74.53% surge over M1). M6 exhibits slight improvement over M5 (macro AUROC 0.6455, 155,888 FP), but underperforms relative to the baseline single models under the evaluated training protocol.



(a) Macro ROC curves comparison

(b) Macro Precision–Recall curves comparison

Figure 3: Overall test-set comparison of (a) Macro-averaged Receiver Operating Characteristic (ROC) curves and (b) Macro-averaged Precision–Recall (PR) curves across models M1 through M6 on NIH ChestX-ray14.

5. **Operating-Point Effect of M7:** When M6 is evaluated using validation-tuned thresholds (M7), total false positives drop from 155,888 to 95,240 (a 38.9% reduction relative to M6 at 0.50, and 7.06% lower than M1 at 0.50). However, this operating-point shift causes recall to drop from 0.6604 to 0.4540 (a 31.3% relative loss in sensitivity). M7's apparent FP reduction relative to M1 at $\tau = 0.50$ is primarily an operating-point effect rather than evidence that the underlying M6 representation is competitive with M1 or M3.

4.2 ROC and Precision–Recall Analysis

Figures 3a and 3b illustrate the macro-averaged Receiver Operating Characteristic and Precision–Recall curves across all models.

The macro ROC curves confirm that M3 maintains a consistent true positive rate advantage across the low false positive rate regime ($0.0 \leq \text{FPR} \leq 0.3$), which corresponds to the low-FPR operating region examined in this study. In the Precision–Recall curves, the separation between the upper tier (M1, M2, M3, M4) and the cross-attention models (M5, M6) is pronounced. Because the average positive class prevalence is low ($\sim 7.14\%$), the macro AUPRC values are concentrated between 0.25 and 0.28 for competent models, while dropping to ~ 0.11 for M5 and M6.

Figure 4 and Figure 5 provide the complete 14-disease breakdown of ROC and PR curves across all candidate architectures.

4.3 False-Positive and Error Dynamics

Figure 6 visualizes total false positive counts and relative percentage reductions across models.

Figure 7 illustrates the per-disease distribution of false positives and the relative change relative to M1. For M3, false positives decreased in 10 out of 14 disease categories, including major reductions in Pneumothorax (19.25% reduction, from 6,765 to 5,463 FP), Infiltration (12.84% reduction, from 12,376 to 10,787 FP), and Emphysema (11.55% reduction, from 3,248 to 2,873 FP). False positives increased modestly for M3 in only 4 categories: Fibrosis (+224 FP), Pleural Thickening (+1,204 FP), Cardiomegaly (+186 FP), and Hernia (+9 FP).

While false positives decreased markedly, diagnostic error analysis requires examining false negatives concurrently. For M3, total false negatives across the 25,596 test radiographs increased

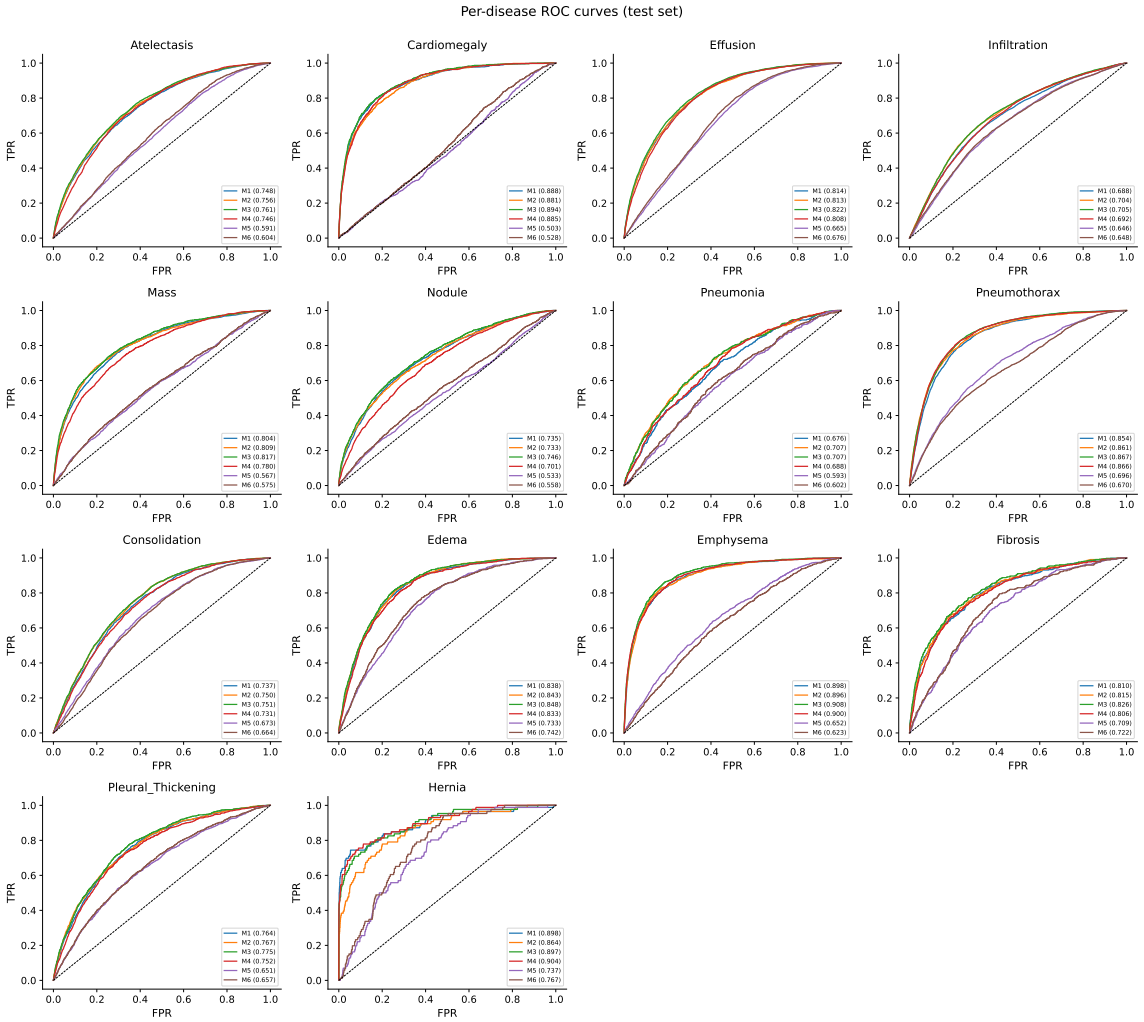


Figure 4: Per-class Receiver Operating Characteristic (ROC) curves for each of the 14 thoracic disease categories across all models (M1–M6) on the 25,596 test radiographs.

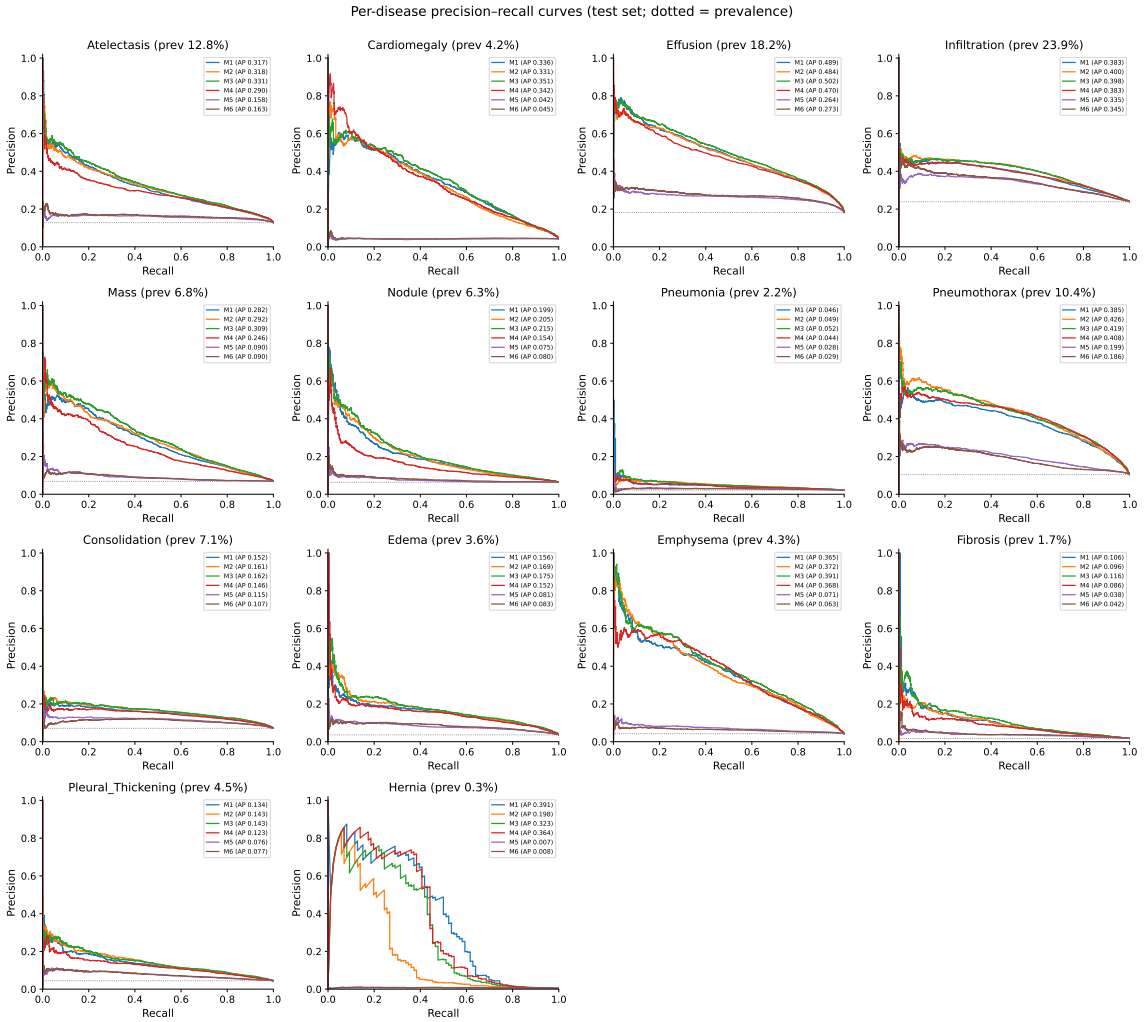
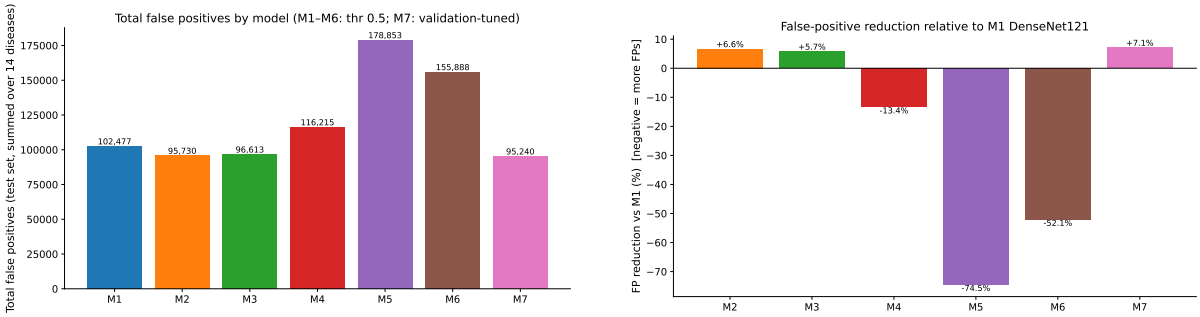


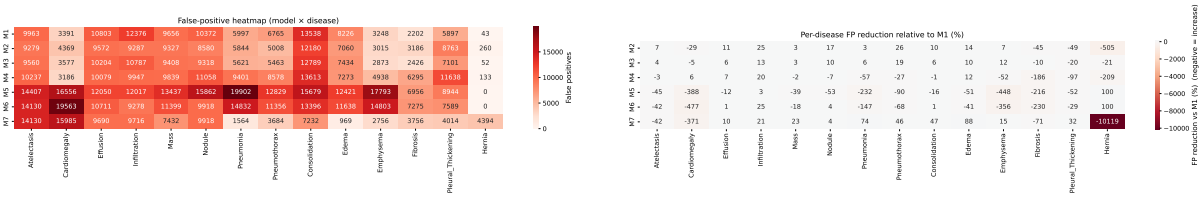
Figure 5: Per-class Precision–Recall (PR) curves for each of the 14 thoracic disease categories across all models (M1–M6) on the 25,596 test radiographs.



(a) Total False Positives across test cohort

(b) Percentage FP reduction vs. M1 baseline

Figure 6: Comparison of diagnostic false-positive behavior: (a) Total test-set false positives across all 14 conditions, and (b) Percentage reduction in false-positive errors relative to the DenseNet-121 (M1) baseline.



(a) Heatmap of absolute FP counts

(b) Heatmap of FP reduction (%) vs. M1

Figure 7: Per-disease diagnostic false-positive error heatmaps: (a) Absolute number of false positives per disease category, and (b) Relative false-positive percentage change versus M1.

by 111 cases relative to M1 (from 4,881 to 4,992, a +2.27% relative change). Net diagnostic classification errors (Total FP + Total FN) decreased from 107,358 for M1 to 101,605 for M3, yielding an overall reduction of 5,753 erroneous decisions across the 358,344 total diagnostic evaluations. Across the 14 individual conditions, false negatives decreased for M3 in 6 pathologies (Atelectasis: 646 vs. 672; Cardiomegaly: 249 vs. 272; Mass: 286 vs. 291; Pneumonia: 281 vs. 297; Emphysema: 250 vs. 262; Fibrosis: 205 vs. 232), but increased in 8 pathologies (most notably Infiltration: 1,100 vs. 925; Pneumothorax: 449 vs. 391; and Nodule: 434 vs. 390). In comparison, Swin-T (M2) incurred an even larger increase of 833 false negatives (5,714 vs. 4,881), whereas feature concatenation (M4) reduced false negatives to 4,753 at the expense of adding 13,738 false positives. Thus, under the standard fixed 0.50 operating threshold, late fusion shifts the operational balance toward higher specificity, achieving fewer false alarms while incurring a modest, localized increase in missed findings.

4.4 Per-Disease Diagnostic Performance

Tables 5 and 6 present the per-disease AUROC and F1 scores across all 14 conditions.

As shown in Table 5, M3 achieves the highest AUROC in 12 out of 14 pathologies, with M2 outperforming on Pneumonia (0.7071 vs. 0.7066) and M4 outperforming on Hernia (0.9035 vs. 0.8973). In Table 6, M3 achieves higher F1 scores than M1 across 11 of 14 diseases.

Table 7 details the corresponding counts of false positives and false negatives for each condition.

4.5 Statistical Significance and Uncertainty Analysis

To establish whether observed performance differences represent robust algorithmic effects rather than finite-sample variability, we evaluated the held-out test predictions under the cluster-

Table 5: Per-disease AUROC scores across all 14 conditions on the 25,596 test radiographs. Highest score per disease among standard architectures (M1–M6) is highlighted in bold.

Disease Category	M1	M2	M3	M4	M5	M6	M7*
Atelectasis	0.7481	0.7556	**0.7610**	0.7458	0.5914	0.6038	0.6038
Cardiomegaly	0.8881	0.8808	**0.8942**	0.8853	0.5025	0.5282	0.5282
Effusion	0.8136	0.8134	**0.8216**	0.8083	0.6650	0.6764	0.6764
Infiltration	0.6882	0.7038	**0.7054**	0.6923	0.6462	0.6481	0.6481
Mass	0.8044	0.8092	**0.8169**	0.7801	0.5673	0.5750	0.5750
Nodule	0.7348	0.7328	**0.7458**	0.7009	0.5329	0.5576	0.5576
Pneumonia	0.6759	**0.7071**	0.7066	0.6885	0.5933	0.6024	0.6024
Pneumothorax	0.8537	0.8609	**0.8673**	0.8657	0.6962	0.6701	0.6701
Consolidation	0.7371	0.7501	**0.7515**	0.7308	0.6733	0.6643	0.6643
Edema	0.8376	0.8434	**0.8480**	0.8325	0.7333	0.7424	0.7424
Emphysema	0.8975	0.8963	**0.9085**	0.9003	0.6516	0.6226	0.6226
Fibrosis	0.8103	0.8150	**0.8261**	0.8060	0.7085	0.7219	0.7219
Pleural Thickening	0.7636	0.7670	**0.7750**	0.7516	0.6506	0.6569	0.6569
Hernia	0.8981	0.8644	0.8973	**0.9035**	0.7366	0.7674	0.7674
Macro Mean	0.7965	0.8000	**0.8089**	0.7923	0.6392	0.6455	0.6455

*M7 reflects identical probabilistic predictions to M6; AUROC is threshold-invariant.

aware inferential framework formalized in Section 3.8. Table 8 presents the inferential statistics for the primary comparison between Late Probability Fusion (M3) and the DenseNet-121 baseline (M1) across $B = 2,000$ patient-cluster bootstrap replicates.

4.5.1 Primary Endpoint Evaluation (M3 vs. M1)

Under the primary evaluation family of four core endpoints with step-down Holm–Bonferroni correction, Late Probability Fusion (M3) demonstrates statistically significant improvements over the DenseNet-121 baseline (M1) across all four primary measures:

1. **Macro AUROC:** Increases by +0.0124 (95% CI: $[+0.0104, +0.0147]$, cluster $z = 11.54$, Wald $p < 10^{-20}$, shifted bootstrap $p < 0.001$, $p_{\text{Holm}} < 0.0001$). The 95% confidence interval is strictly positive and bounded well above zero.

2. **Macro Average Precision:** Increases by +0.0103 (95% CI: $[+0.0066, +0.0144]$, cluster $z = 5.19$, Wald $p = 2.1 \times 10^{-7}$, shifted bootstrap $p < 0.001$, $p_{\text{Holm}} = 4.2 \times 10^{-7}$).

3. **Macro F1 at Standard Operating Point ($\tau = 0.50$):** Improves by +0.0073 (95% CI: $[+0.0030, +0.0112]$, cluster $z = 3.49$, Wald $p = 0.00048$, shifted bootstrap $p = 0.0010$, $p_{\text{Holm}} = 0.00048$).

4. **Total False-Positive Reduction:** Total false-positive decisions decrease by $-5,864$ (95% CI: $[-6,648, -5,131]$, cluster $z = -14.95$, Wald $p < 10^{-40}$, shifted bootstrap $p < 0.001$, $p_{\text{Holm}} < 0.0001$), representing a relative reduction of 5.72% (95% CI: [5.10%, 6.41%]).

4.5.2 Secondary Endpoints and Dual-Inference False-Negative Dynamics

While secondary aggregate metrics including micro F1 ($\Delta = +0.0105$, $p < 10^{-35}$) and specificity ($\Delta = +0.0191$, $p < 10^{-40}$) confirm significant overall gains, macro precision alone does not attain significance ($\Delta = +0.0040$, 95% CI: $[-0.0022, +0.0097]$, cluster $z = 1.31$, $p = 0.1899$),

Table 6: Per-disease F1 scores across all 14 conditions on the 25,596 test radiographs. Highest score per disease among standard architectures (M1–M6) is highlighted in bold.

Disease Category	**M1**	**M2**	**M3**	**M4**	**M5**	**M6**	**M7**
Atelectasis	0.3290	0.3388	**0.3404**	0.3280	0.2538	0.2581	0.2581
Cardiomegaly	**0.3032**	0.2612	0.3000	0.3021	0.0793	0.0857	0.0866
Effusion	0.4345	**0.4519**	0.4473	0.4464	0.3803	0.3900	0.3891
Infiltration	0.4382	**0.4679**	0.4575	0.4620	0.4287	0.4314	0.4311
Mass	0.2266	0.2301	**0.2317**	0.2157	0.1381	0.1441	0.1483
Nodule	0.1864	**0.1966**	0.1960	0.1726	0.1172	0.1303	0.1303
Pneumonia	0.0758	**0.0861**	0.0850	0.0692	0.0476	0.0524	0.0452
Pneumothorax	0.3886	**0.4285**	**0.4285**	0.3562	0.2444	0.2352	0.2738
Consolidation	0.1923	**0.2035**	0.1995	0.1889	0.1685	0.1766	0.1922
Edema	0.1602	**0.1761**	0.1715	0.1691	0.1122	0.1162	0.1032
Emphysema	0.3213	0.3382	**0.3506**	0.2665	0.0984	0.1005	0.1058
Fibrosis	0.1430	0.1263	**0.1488**	0.0883	0.0649	0.0682	0.0678
Pleural Thickening	**0.1849**	0.1592	0.1799	0.1382	0.1256	0.1300	0.1420
Hernia	**0.4734**	0.1693	0.4229	0.3218	0.0000	0.0000	0.0186
Macro Mean	0.2755	0.2596	**0.2828**	0.2518	0.1614	0.1656	0.1709

and macro recall improvement remains marginal ($\Delta = +0.0059$, 95% CI: $[-0.0004, +0.0119]$, cluster $z = 1.89$, $p = 0.0588$).

Crucially, the evaluation of Total False Negatives (FN) reveals an important methodological distinction between asymptotic and bootstrap inference procedures. The point estimate indicates 111 additional false negatives for M3 compared to M1 (4,992 vs. 4,881, a +2.27% relative change). Under the unadjusted asymptotic cluster-Wald test, this difference yields $z = 1.991, p = 0.0465$, with the asymptotic normal 95% confidence interval $[+1.73, +220.27]$ narrowly excluding zero, while the shifted empirical bootstrap test centered under H_0 yields $p_{\text{shift}} = 0.0445$ (88 of 2,000 centered resamples with $|\widetilde{\Delta}^*| \geq 111$). In contrast, under the non-parametric patient-cluster bootstrap ($B = 2,000$), the empirical 95% percentile confidence interval is $[-1, +217]$, which includes zero because 53 of the 2,000 resamples (2.65% > 2.5%) yield differences ≤ 0, corresponding to an uncentered two-sided percentile tail probability of $p = 0.0535$. Furthermore, Total FN is treated as an exploratory secondary endpoint rather than one of the four primary family outcomes; when evaluated under family-wise multiplicity adjustment across secondary metrics, it is non-significant ($p_{\text{adj}} \geq 0.093$).

Consequently, we preserve this dual-inference distinction and do not claim that the false-negative increase is either definitively significant or definitively absent. The empirical finding indicates that under the fixed $\tau = 0.50$ threshold, late fusion reflects an operational trade-off toward higher specificity rather than an unconditional reduction across both error modes.

4.5.3 Per-Disease Multiplicity Control Across 14 Pathologies

At the individual pathology level, 13 of the 14 disease categories demonstrate positive AUROC point estimates for M3 over M1. Following step-down Holm-Bonferroni correction across the 14 simultaneous hypothesis tests, all 13 improvements remain statistically significant ($p_{\text{Holm}} \leq 0.0165$, and all 13 satisfy Benjamini-Hochberg FDR $q < 0.01$). The largest significant AUROC gains occur in Pneumonia ($\Delta = +0.0306$, $p_{\text{Holm}} = 1.31 \times 10^{-7}$), Infiltration ($\Delta = +0.0172$, $p_{\text{Holm}} < 10^{-15}$), and Fibrosis ($\Delta = +0.0157$, $p_{\text{Holm}} = 0.0125$). For Hernia, which contains only 86 positive test cases (0.34% prevalence), the AUROC difference is negligible ($\Delta = -0.0008$, 95% CI: $[-0.0167, +0.0206]$, $p_{\text{Holm}} = 0.9355$), confirming lack of statistical evidence to differentiate

the models on rare findings.

4.5.4 Secondary Pairwise Model Comparisons

Holm-corrected comparisons across the candidate architectures confirm key architectural trade-offs:

- **M2 (Swin-T) vs. M1 (DenseNet-121):** M2 does not differ significantly from M1 in macro AUROC ($\Delta = +0.0035$, 95% CI: $[-0.0003, +0.0072]$, $p_{\text{Holm}} = 0.1336$) or macro AP ($\Delta = -0.0070$, $p_{\text{Holm}} = 0.0617$), but incurs significantly lower macro F1 ($\Delta = -0.0160$, $p_{\text{Holm}} < 0.0001$) and generates 833 additional false negatives.
- **M4 (Feature Concat) vs. M1:** M4 underperforms M1 significantly in macro AP ($\Delta = -0.0119$, $p_{\text{Holm}} = 0.0013$) and macro F1 ($\Delta = -0.0237$, $p_{\text{Holm}} < 0.0001$), while generating a statistically significant increase of $+13,738$ false positives ($p_{\text{Holm}} < 0.0001$).
- **Cross-Attention Models (M5 and M6) vs. M1:** Cross-attention models exhibit statistically established degradation across all primary metrics ($\Delta_{\text{AUROC}} \leq -0.1510$, $p_{\text{Holm}} < 0.0001$), accompanied by massive false-positive inflation ($+76,376$ FP for M5 and $+53,411$ FP for M6, both $p_{\text{Holm}} < 0.0001$).
- **M3 vs. All Alternative Architectures:** Late Probability Fusion (M3) significantly outperforms all alternative architectures (M2, M4, M5, M6) in both macro AUROC and macro AP ($p_{\text{Holm}} < 0.0001$ for all pairwise tests).

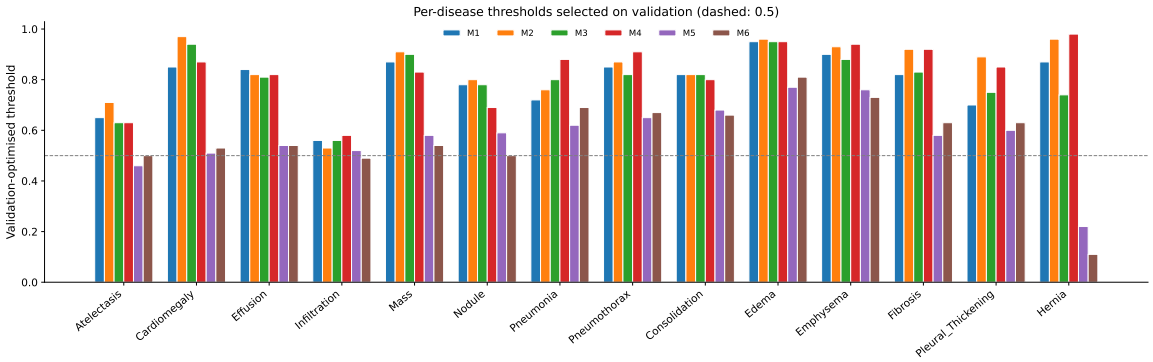
4.6 Threshold and Operating-Point Analysis

Figure 8a displays the validation-tuned threshold values τ_c^* across all 14 diseases. For high-prevalence conditions like Atelectasis and Infiltration, the optimal thresholds hover close to 0.50 (0.50 and 0.49). For rare conditions like Hernia, the threshold shifts to 0.11, while for conditions with high baseline false positives like Edema and Emphysema, the thresholds rise to 0.81 and 0.73.

To determine whether the false-positive reduction observed in M7 is an architectural property of M6 or an artifact of threshold adjustment, we evaluate all models under their own validation-tuned thresholds. Table 9 presents this like-for-like comparison.

This like-for-like analysis provides three insights:

1. When DenseNet-121 (M1) is evaluated with its own validation-tuned thresholds, total false positives fall from 102,477 to 38,409 (a 62.52% reduction), macro precision reaches 0.2809, macro recall is 0.4431, macro F1 rises to 0.3286, and micro F1 rises to 0.3772.
2. When other models are tuned on validation thresholds, Swin-T (M2) achieves 30,940 false positives and macro F1 of 0.3227; Late Fusion (M3) achieves 32,797 false positives (68.00% reduction vs. M1 at 0.50), macro F1 of 0.3368, and micro F1 of 0.3983; and feature concatenation (M4) achieves 33,387 false positives and macro F1 of 0.3187.
3. In contrast, cross-attention models under validation-tuned thresholds produce 100,316 false positives with macro F1 of 0.1725 for M5, and 95,240 false positives with macro precision of 0.1147, macro recall of 0.4540, and macro F1 of 0.1709 (micro F1 = 0.2263) for M6/M7. Thus, M7's apparent FP reduction relative to M1 at $\tau = 0.50$ is primarily an operating-point effect rather than evidence that the underlying M6 representation is competitive with M1 or M3. M7 represents an operating-point analysis rather than an improved architecture.



(a) Validation-optimized thresholds per disease



(b) Confusion matrices for M7

Figure 8: (a) Disease-specific classification thresholds τ_c^* optimized on validation data. (b) Diagnostic confusion matrices for M7 across the 14 conditions under validation-optimized operating points.

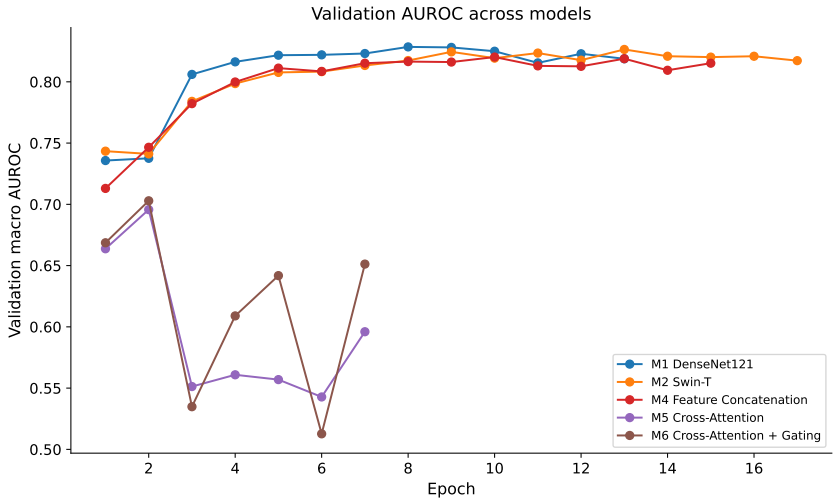


Figure 9: Validation macro AUROC across training epochs for M1, M2, M4, M5, and M6. Note the substantial performance degradation of M5 and M6 during Stage 2 backbone unfreezing (epoch 3 onward), reflecting observed optimization instability under the evaluated training protocol and triggering early stopping at epoch 7.

4.7 Training Dynamics and Failure Analysis

Figure 9 depicts the validation AUROC trajectories across epochs for all trained models. Table 10 summarizes the training time, epoch history, and inference latencies.

A closer inspection of the training logs reveals the mechanical basis for the underperformance of M5 and M6:

1. **Stage 1 Behavior:** During the first two epochs (warm-up with frozen backbones), M5 and M6 achieved reasonable validation AUROCs of 0.6957 and 0.7029.

2. **Stage 2 Optimization Instability:** At epoch 3, when both the DenseNet and Swin backbones were unfrozen for end-to-end training at $\eta = 1.0\times10^{-4}$, substantial performance degradation occurred: validation AUROC dropped immediately to 0.5513 (M5) and 0.5348 (M6), with validation loss increasing from 1.365 to 1.711. The observed degradation coincided with the transition to end-to-end backbone fine-tuning. Possible contributing factors include simultaneous adaptation of both pretrained backbones and the relatively high-dimensional cross-attention fusion module.

3. **Failure to Recover During Fine-Tuning:** Over the subsequent four fine-tuning epochs (epochs 4–7), the validation AUROC recovered only marginally (to 0.5960 for M5 and 0.6512 for M6). Because early stopping patience was set to 5 epochs and none of the fine-tuning epochs exceeded the Stage 1 peak at epoch 2, early stopping triggered at epoch 7. Consequently, the best saved checkpoints restored for final test evaluation were those from Epoch 2, which had received only 2 warm-up epochs on the head while their backbones retained generic ImageNet initialization.

4. **Absence of Domain Pre-training:** In our verified configuration (`HYBRID_INIT_FROM_M1_M2` = False), backbones for M5 and M6 were initialized directly from ImageNet-1K checkpoints, rather than from domain fine-tuned M1 and M2 weights. Propagating cross-attention gradients simultaneously into two heterogeneous unadapted backbones under a single learning rate schedule may have contributed to optimization instability through interactions between gradients from the two jointly fine-tuned backbones.

Table 7: Per-disease counts of diagnostic False Positives (FP) and False Negatives (FN) across all models on the 25,596 test cases.

Disease	M1		M2		M3		M4		M5		M6		M7	
	FP	FN	FP	FN	FP	FN	FP	FN	FP	FN	FP	FN	FP	FN
Atelectasis	9,963	672	9,279	718	9,560	646	10,237	628	14,407	708	14,130	699	14,130	699
Cardiomegaly	3,391	272	4,369	252	3,577	249	3,186	312	16,556	341	19,563	145	15,985	297
Effusion	10,803	367	9,572	504	10,204	377	10,079	424	12,050	735	10,711	935	9,690	1,192
Infiltration	12,376	925	9,287	1,409	10,787	1,100	9,947	1,288	12,017	1,166	9,278	1,879	9,716	1,763
Mass	9,656	291	9,327	308	9,408	286	9,839	347	13,437	622	11,399	727	7,432	1,013
Nodule	10,372	390	8,580	511	9,318	434	11,058	425	15,862	535	9,918	819	9,918	819
Pneumonia	5,997	297	5,844	267	5,621	281	9,401	198	19,902	56	14,832	141	1,564	506
Pneumothorax	6,765	391	5,008	573	5,463	449	8,578	229	12,829	508	11,356	796	3,684	1,658
Consolidation	13,538	182	12,180	230	12,789	197	13,613	206	15,679	206	13,396	342	7,232	853
Edema	8,226	128	7,060	154	7,434	141	7,273	168	12,421	132	11,638	150	969	822
Emphysema	3,248	262	3,015	257	2,873	250	4,938	166	17,793	116	14,803	252	2,756	878
Fibrosis	2,202	232	3,186	191	2,426	205	6,295	124	6,956	187	7,275	163	3,756	288
Pleural Thickening	5,897	426	8,763	286	7,101	328	11,638	194	8,944	467	7,589	536	4,014	749
Hernia	43	46	260	54	52	49	133	44	0	86	0	86	4,394	44
Total Sum	**102,477**	**4,881**	**95,730**	**5,714**	**96,613**	**4,992**	**116,215**	**4,753**	**178,853**	**5,865**	**155,888**	**7,670**	**95,240**	**11,581**

Table 8: Inferential statistical evaluation for Late Probability Fusion (M3) versus DenseNet-121 Baseline (M1) on the held-out ChestX-ray14 test set ($N = 25,596$ radiographs, $K = 2,797$ patient clusters, $B = 2,000$ cluster bootstrap replicates). Primary family endpoints are controlled for family-wise error rate using the step-down Holm-Bonferroni procedure at $\alpha = 0.05$.

Metric	Primary Family?	M1 Point	M3 Point	Paired Diff	Cluster SE	95% Cluster CI	Cluster z	Wald p	Holm p	Significant?
Macro AUROC	**YES**	0.7965	0.8089	**+0.0124**	0.0011	[+0.0104, +0.0147]	11.54	$< 10^{-20}$	< 0.0001	**YES**
Macro AP (AUPRC)	**YES**	0.2673	0.2776	**+0.0103**	0.0020	[+0.0066, +0.0144]	5.19	2.1×10^{-7}	4.2×10^{-7}	**YES**
Macro F1 ($\tau = 0.50$)	**YES**	0.2755	0.2828	**+0.0073**	0.0021	[+0.0030, +0.0112]	3.49	0.00048	**0.00048**	**YES**
Total False Positives	**YES**	102,477	96,613	**−5,864**	392.3	[−6,648, −5,131]	−14.95	$< 10^{-40}$	< 0.0001	**YES**
Relative FP Red. (%)	NO	0.0%	5.72%	**+5.72%**	0.35%	[+5.10%, +6.41%]	16.55	$< 10^{-40}$	—	**YES**
Micro F1	NO	0.2937	0.3042	**+0.0105**	0.0008	[+0.0090, +0.0120]	13.72	$< 10^{-35}$	—	**YES**
Specificity	NO	0.6805	0.6996	**+0.0191**	0.0011	[+0.0170, +0.0213]	17.55	$< 10^{-40}$	—	**YES**
False Positive Rate	NO	0.3195	0.3004	**−0.0191**	0.0011	[−0.0213, −0.0170]	−17.55	$< 10^{-40}$	—	**YES**
Macro Precision	NO	0.1844	0.1884	+0.0040	0.0030	[−0.0022, +0.0097]	1.31	0.1899	—	NO
Macro Recall	NO	0.7359	0.7418	+0.0059	0.0031	[−0.0004, +0.0119]	1.89	0.0588	—	NO
Total False Negatives*	NO	4,881	4,992	+111	55.8	[−1, +217]	1.99	0.0465	—	Inconclusive
Relative FN Change (%)	NO	0.0%	+2.27%	+2.27%	1.16%	[−0.02%, +4.46%]	1.97	0.0491	—	Inconclusive

*Total False Negatives exhibits method-dependent inference: unadjusted asymptotic cluster-Wald test yields $z = 1.991, p = 0.0465$ with asymptotic Normal 95% CI [+1.73, +220.27] (narrowly excluding zero), and the shifted empirical bootstrap test under H_0 yields $p_{\text{shift}} = 0.0445$. In contrast, the non-parametric patient-cluster bootstrap 95% percentile CI is [−1, +217], which includes zero because 53 of 2,000 resamples are ≤ 0 (corresponding to an uncentered two-sided percentile tail probability of $p = 0.0535$). Total FN is an exploratory secondary outcome and is non-significant under multiplicity control ($p_{\text{adj}} \geq 0.093$).

Table 9: Supplementary like-for-like comparison: Every architecture evaluated using its own frozen, validation-optimized disease-specific thresholds τ_c^*. Relative FP reduction is computed versus M1 evaluated at standard $\tau = 0.50$.

Model [Val-Tuned]	Macro AUROC	Macro AP	Macro Prec.	Macro Recall	Macro F1	Micro F1	Macro FPR	Total FP	Hamming Loss	FP Red. vs M1 (0.5)
M1 DenseNet121	0.7965	0.2673	0.2809	0.4431	0.3286	0.3772	0.1239	38,409	0.1405	+62.52%
M2 Swin-T	0.8000	0.2602	0.2901	0.3966	0.3227	0.3890	0.1000	30,940	0.1231	+69.81%
M3 Late Fusion	**0.8089**	**0.2776**	**0.3084**	0.4182	**0.3368**	**0.3983**	0.1063	32,797	0.1258	+68.00%
M4 Feat. Concat	0.7923	0.2554	0.2808	0.4044	0.3187	0.3721	0.1069	33,387	0.1304	+67.42%
M5 Cross-Attn	0.6392	0.1128	0.1136	**0.4591**	0.1725	0.2258	0.3117	100,316	0.3106	+2.11%
M6 / M7 Gated	0.6455	0.1136	0.1147	0.4540	0.1709	0.2263	0.2955	95,240	0.2981	+7.06%

Table 10: Computational complexity, training duration, and inference throughput. Training time denotes wall-clock training duration. Pure inference latency is measured on GPU with batch size 32 using CUDA synchronization without I/O; inference throughput measurements include the full test-set disk-loading time.

Model	**Total Params**	**Train Time (min)**	**Epochs Run**	**Best Epoch**	**Best Val AUROC**	**Pure Model ms/image**	**Pure img/s**
M1 DenseNet-121	6,968,206	226.74	13	8	0.8285	1.721	580.97
M2 Swin-T	27,530,120	354.49	17	13	0.8264	1.609	621.38
M3 Late Fusion	34,498,326	—	—	—	0.8375*	3.383	295.61
M4 Feat. Concat	35,925,768	438.60	15	10	0.8202	3.159	316.60
M5 Cross-Attn	37,554,952	205.08	7	2	0.6957	3.568	280.27
M6 Gated Attn	41,233,160	271.63	7	2	0.7029	3.490	286.52

*M3 value represents validation macro AUROC obtained at retained $\alpha = 0.50$ (0.8375; validation sweep shallow optimum reached 0.8376 at $\alpha = 0.55$–0.60) without additional gradient training.

5 Discussion

The empirical findings from this study address core questions regarding architectural design for multi-label medical image interpretation:

1. Which model performed best? Across the evaluated configurations, Late Probability Fusion (M3) between DenseNet-121 and Swin-T achieved the strongest aggregate performance among the evaluated architectures under the standard $\tau = 0.50$ operating point, achieving higher macro AUROC (0.8089), macro AP (0.2776), macro precision (0.1884), macro F1 (0.2828), and micro F1 (0.3042). While M4 achieved higher recall (0.7785) and M2 attained higher specificity (0.7039), M3 maintained the most balanced trade-off across both ranking and operating-point metrics.

2. Did late fusion help? Yes. Combining predictions from models trained with divergent inductive biases provided a consistent advantage over both individual constituent models. DenseNet-121 relies on translation-equivariant local receptive fields, whereas Swin-T leverages shifted-window self-attention. A validation sweep identified a shallow optimum around $\alpha = 0.55$–0.60 (validation macro AUROC 0.8376); $\alpha = 0.50$ (validation macro AUROC 0.8375) was retained as the final fusion weight for test evaluation. This balanced weighting improved validation AUROC compared to single models (0.8231 for pure Swin-T at $\alpha = 0.00$ and 0.8295 for pure DenseNet-121 at $\alpha = 1.00$), and translated to a statistically established 5.72% reduction in false positives on the unseen test set ($p_{\text{Holm}} < 0.0001$) alongside higher macro recall (0.7418 vs. 0.7359) and F1. Concurrently, total false negatives exhibited a modest and statistically inconclusive increase of 111 cases (4,992 vs. 4,881; 95% CI: $[-1, +217]$).

3. Did feature concatenation help? Intermediate feature concatenation (M4) succeeded in expanding sensitivity, achieving the highest recall (0.7785) among all models. However, it failed to improve overall discriminative precision, resulting in 13,738 additional false positives (+13.41% vs. M1) and lower macro AUROC (0.7923). In high-volume screening settings where false alarms impose significant clinical burdens, this trade-off may be disadvantageous.

4. Did cross-attention and gating help? In the current implementation, cross-attention (M5) and disease-specific gating (M6) did not yield competitive performance. While disease-specific gating (M6) improved slightly upon standard cross-attention (M5) by raising AUROC from 0.6392 to 0.6455 and reducing false positives by 22,965, both models fell substantially short of the single-backbone baselines.

5. What does the negative result of M5 and M6 reveal? This negative finding is scientifically instructive. In literature, cross-attention is frequently presumed to offer higher expressive capacity. However, under the evaluated training protocol, M5 and M6 performed substantially worse than the simpler baselines. The observed degradation coincided with the transition to end-to-end backbone fine-tuning. Possible contributing factors include simultaneous adaptation of both pretrained backbones, the relatively high-dimensional cross-attention fusion module, and initialization from ImageNet rather than domain-adapted weights. Highly parameterized interaction mechanisms may require specialized training strategies—such as staged unfreezing, lower backbone learning rates, or domain-adapted initialization. Crucially, the observation that M5 and M6 performed poorly under this specific protocol does not establish that cross-attention itself is inherently inferior, but rather underscores observed optimization instability under the evaluated training regime.

6. What happened to false positives versus recall and false negatives? Our results demonstrate that diagnostic error reduction involves distinct, model-dependent trade-offs across error modes. At the fixed $\tau = 0.50$ threshold, Swin-T (M2) reduced false positives by 6.58% relative to DenseNet-121 (from 102,477 to 95,730 FP), but this occurred at the direct expense of reduced recall (0.7267 vs. 0.7359), lower macro F1 (0.2596 vs. 0.2755), and 833 additional false negatives (5,714 vs. 4,881). Conversely, feature concatenation (M4) expanded sensitivity (recall 0.7785, 4,753 FN) at the direct cost of a 13.41% increase in false positives (116,215 vs. 102,477 FP). Late probability fusion (M3) achieved the most favorable overall balance, significantly reducing false positives by 5.72% ($-5,864$ FP, $p_{\text{Holm}} < 0.0001$) while increasing macro recall (0.7418 vs. 0.7359), macro AUROC, and macro AP. However, M3 also produced 111 more false negatives than M1 (4,992 vs. 4,881; 95% CI: $[-1, +217]$). Thus, late fusion does not achieve an unconditional decrease in both error modes simultaneously, but rather reflects an operational trade-off toward higher specificity under the evaluated threshold.

7. How sensitive are conclusions to threshold selection? Threshold selection exerts a profound influence on operational performance. The evaluation of M7 demonstrated that tuning thresholds on validation data can cut false positives substantially, but can also severely depress recall. Crucially, when baseline models M1–M4 were evaluated under an identical validation-tuning protocol, their false positives dropped by over 60% while maintaining substantially higher F1 scores than M7. This underscores the necessity of like-for-like thresholding protocols when evaluating architectural innovations.

6 Limitations

Several limitations of this study must be acknowledged:

1. **Single Retrospective Dataset:** All experiments were conducted on the NIH ChestX-ray14 dataset. While large and standardized, external generalization to other benchmark repositories (e.g., CheXpert, MIMIC-CXR) or prospective clinical populations was not evaluated.

2. **Noisy NLP-Derived Labels:** Ground truth labels in ChestX-ray14 were extracted from unstructured radiology reports via automated natural language processing, introducing known label noise and ambiguity.

3. **Suboptimal Training Regime for Cross-Attention:** Due to computational constraints and early stopping triggering at epoch 7, models M5 and M6 were evaluated at checkpoints that had not converged to optimal fine-tuned representations. Their underperformance reflects observed optimization instability under the evaluated training protocol rather than establishing that cross-attention is intrinsically inferior.

4. **Single-Dataset Retrospective Inference and Generalization:** Although we implemented rigorous patient-cluster bootstrap and cluster-robust Wald testing to account for intra-patient correlation across all 25,596 test radiographs, our statistical inferences are strictly retrospective and derived from a single benchmark dataset. The primary evaluation family and hypothesis structures were formalized as part of this comprehensive retrospective audit rather than pre-registered prospectively prior to model training. External validation on multi-center cohorts (e.g., CheXpert or MIMIC-CXR) is required to confirm whether the observed statistical advantages of decision-level late fusion generalize across distinct clinical settings, imaging protocols, scanner manufacturers, and patient populations.

5. **No Prospective Clinical Validation:** The reported diagnostic error reductions reflect algorithmic properties on benchmark data and cannot be construed as evidence of clinical efficacy or readiness for diagnostic deployment.

7 Conclusion

In this empirical comparative study, we systematically evaluated six deep learning architectures and seven experimental configurations on the NIH ChestX-ray14 benchmark to investigate how CNN–Transformer integration strategies govern multi-label classification accuracy and diagnostic error behavior. Under strict experimental controls and verified patient-level splitting, decision-level Late Probability Fusion (M3) emerged as the most effective strategy among the evaluated architectures, achieving a macro AUROC of 0.8089 (+0.0124 over M1, $p_{\text{Holm}} < 0.0001$), macro AP of 0.2776 ($p_{\text{Holm}} < 0.0001$), macro F1 of 0.2828 ($p_{\text{Holm}} = 0.00048$), and reducing total test-set false positives by 5.72% relative to DenseNet-121 ($p_{\text{Holm}} < 0.0001$). Total false negatives showed a modest and statistically inconclusive increase of 111 cases (4,992 vs. 4,881, 95% CI: $[-1, +217]$), reflecting an operational trade-off toward higher specificity under the standard 0.50 operating threshold. In contrast, intermediate feature concatenation increased recall at the expense of a 13.41% increase in false positives, while spatial cross-attention hybrids exhibited substantial performance degradation during end-to-end fine-tuning under the evaluated training protocol. Furthermore, operating-point analysis revealed that threshold adjustment significantly shifts operating-point trade-offs and must be evaluated under like-for-like protocols. These empirical findings underscore that architectural complexity does not guarantee superior classification performance or diagnostic error reduction, and that simple ensembles combining complementary inductive biases remain a strong baseline for medical computer vision.

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